

Model Selection: Part II

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Applied Econometrics

Penalized Regression

Penalized Regression

Suppose we fit a regression model and penalized extra variables all in one go, what would that look like?

$$\hat{\beta} = \arg \min_{\beta} \left[\frac{1}{2} \sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|^q \right]$$

- ▶ We can consider the penalty term $\lambda \sum_{j=1}^p |\beta_j|^q$ as penalizing models where β gets further away from zero.
- ▶ Similar to placing a **prior distribution** on β_j centered at 0.
- ▶ We definitely want to **standardize** our inputs before using penalized regression methods.
- ▶ Usually you fix q and then look at how estimates respond to γ .
- ▶ There are two famous cases $q = 1$ (Lasso) and $q = 2$ (Ridge) though in practice there are many possibilities.

$$\hat{\beta}^{LASSO} = \arg \min_{\beta} \left[\frac{1}{2} \sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^K x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^K |\beta_j| \right]$$

- ▶ Penalty is L_1 norm on β .
- ▶ Can re-write as a constraint $\sum_{j=1}^K |\beta_j| \leq s$
- ▶ If X is orthonormal then $\hat{\beta}_j^{LASSO} = \text{sign}(\hat{\beta}_j) \cdot (|\hat{\beta}_j| - \lambda)_+$
- ▶ In words: we get coefficients that are closer to zero by λ , but coefficients within λ of zero are shrunk to zero.
- ▶ This leads people to describe LASSO as a **shrinkage** estimator. It produces models that are **sparse**.
- ▶ Instead of a discrete parameter such as the number of lags p we can continuously penalize additional complexity with λ .

But... is choosing λ any easier than choosing p ?

- ▶ We call λ the **regularization** parameter.
- ▶ We can choose λ in a way that minimizes expected prediction error (EPE).
- ▶ Recall $EPE(\lambda) = \mathbb{E}_x \mathbb{E}_{y|x}([Y - g(X, \lambda)]^2 | X)$.
- ▶ In practice most people look at out of sample prediction error rate on a **cross validated sample**.

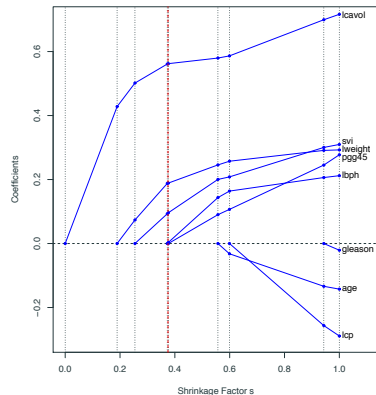


FIGURE 3.10. Profiles of lasso coefficients, as the tuning parameter t is varied. Coefficients are plotted versus $s = t / \sum_1^p |\hat{\beta}_j|$. A vertical line is drawn at $s = 0.36$, the value chosen by cross-validation. Compare Figure 3.8 on page 65; the lasso profiles hit zero, while those for ridge do not. The profiles are piece-wise linear, and so are computed only at the points displayed; see Section 3.4.4 for details.

Another popular alternative is the $q = 2$ case

$$\hat{\beta}^{Ridge} = \arg \min_{\beta} \left[\frac{1}{2} \sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^K x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^K |\beta_j|^2 \right]$$

- ▶ Penalty is L_2 norm on β .
- ▶ Can re-write as a constraint $\sum_{j=1}^K |\beta_j|^2 \leq s$
- ▶ $\hat{\beta}^{Ridge} = (X'X + \lambda I)^{-1} X'Y$.
- ▶ If X is orthonormal then $\hat{\beta}_j^{Ridge} = \hat{\beta}_j / (1 + \lambda)$
- ▶ In words: everything gets dampened by a constant factor λ (we don't get zeros).
- ▶ Adding a constant to the diagonal of $(X'X)$ ensures that the matrix will be invertible even when we have multicollinearity.

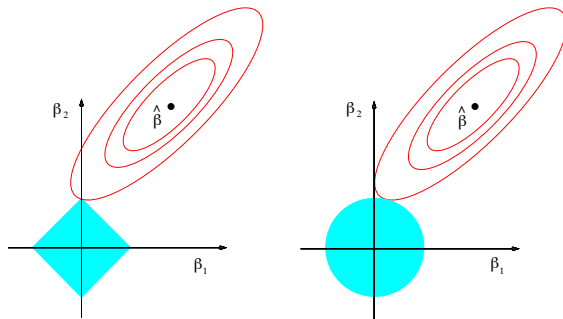


FIGURE 3.11. Estimation picture for the lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \leq t$ and $\beta_1^2 + \beta_2^2 \leq t^2$, respectively, while the red ellipses are the contours of the least squares error function.

What is the point?

Ridge:

- ▶ Ridge doesn't provide sparsity which can be a good thing.
- ▶ It is most helpful (relative to OLS) when X 's are highly correlated with one another.
- ▶ OLS can set large but imprecise coefficients when it cannot disentangle effects.

LASSO:

- ▶ LASSO is useful for variable/feature selection.
- ▶ LASSO does not generally possess the **oracle property** though variants such as **adaptive LASSO** may.
- ▶ LASSO sometimes has the oracle property for $p \gg N$ and cases where the true β 's are not too large.
- ▶ People sometimes use LASSO to choose components and then OLS for unbiased coefficient estimates

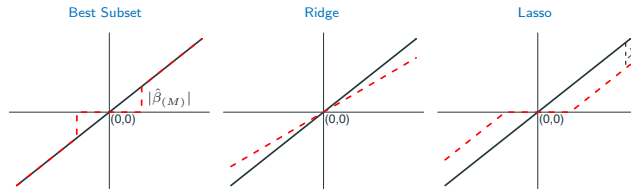
We can actually combine them using **elastic net regression**:

$$P(\lambda_1, \lambda_2, \beta) = \lambda_1 \sum_{j=1}^K |\beta_j| + \lambda_2 \sum_{j=1}^K |\beta_j|^2$$

This includes both the L_1 and L_2 penalties.

TABLE 3.4. Estimators of β_j in the case of orthonormal columns of $\mathbf{X}$. M and λ are constants chosen by the corresponding techniques; sign denotes the sign of its argument (± 1), and x_+ denotes “positive part” of x . Below the table, estimators are shown by broken red lines. The 45° line in gray shows the unrestricted estimate for reference.

Estimator	Formula
Best subset (size M)	$\hat{\beta}_j \cdot I(\hat{\beta}_j \geq \hat{\beta}_{(M)})$
Ridge	$\hat{\beta}_j / (1 + \lambda)$
Lasso	$\text{sign}(\hat{\beta}_j)(\hat{\beta}_j - \lambda)_+$



LAR: Least Angle Regression

Remember Forward Stagewise Regression, consider this alternative:

1. Start with $r = y - \bar{y}$ and $(\beta_1, \dots, \beta_p) = 0$. (Standardize first!)
 2. Find the predictor x_j most correlated with r .
 3. Move β_j from 0 to its least-squares estimate $\langle x_j, r \rangle$ slowly
 4. Update $r \leftarrow r - \delta_j \cdot x_j$.
 5. Keep moving x_j in same direction until x_k has as much correlation with updated r ,
 6. Continue updating (β_j, β_k) in direction of **joint** least-squares coefficients until some other competitor x_l has as much correlation with r .
 7. Continue until all p predictors have entered. After $\min[N - 1, p]$ steps we arrive at full OLS solution.
- **Optional:** If a current least-squares estimate hits zero drop it from the active set and re-estimate the joint least squares direction without it.

Why do we need LAR?

- ▶ It turns out that with the optional step from the previous slide: LAR gives us an easy algorithm to compute the LASSO estimate.
- ▶ Actually it does even better – it gives us the full path of LASSO estimates for all values of λ !
- ▶ This is actually a relatively new result.

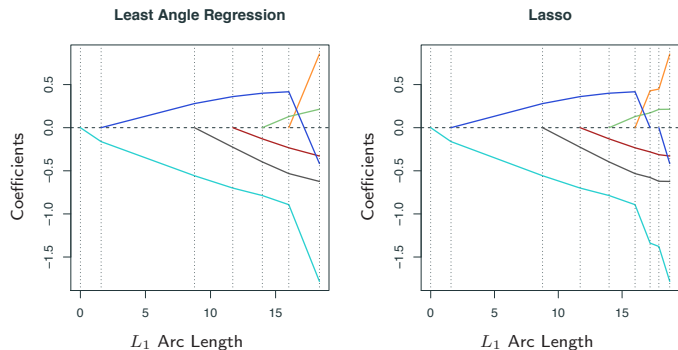
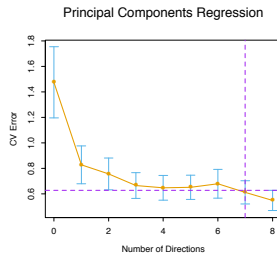
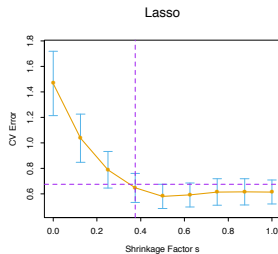
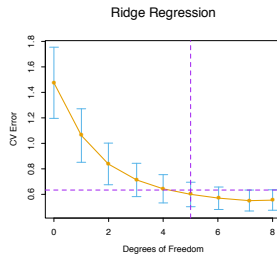
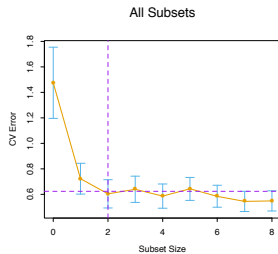


FIGURE 3.15. Left panel shows the LAR coefficient profiles on the simulated data, as a function of the L_1 arc length. The right panel shows the Lasso profile. They are identical until the dark-blue coefficient crosses zero at an arc length of about 18.

Overall Comparison



Overall Comparison

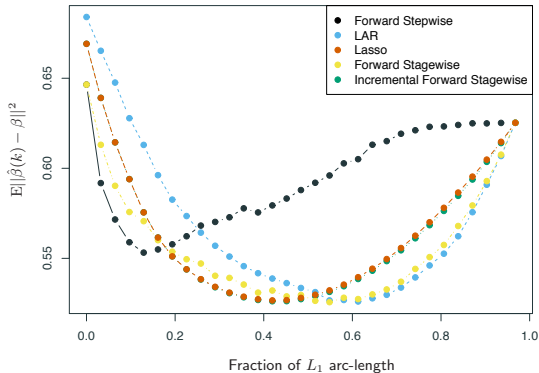


FIGURE 3.16. Comparison of LAR and lasso with forward stepwise, forward stagewise (FS) and incremental forward stagewise (FS_0) regression. The setup is the same as in Figure 3.6, except $N = 100$ here rather than 300. Here the slower FS regression ultimately outperforms forward stepwise. LAR and lasso show similar behavior to FS and FS_0 . Since the procedures take different numbers of steps (across simulation replicates and methods), we plot the MSE as a function of the fraction of total L_1 arc-length toward the least-squares fit.

- ▶ An important question with LASSO is whether or not it produces consistent parameter estimates (Generally **no**).
- ▶ We think of asymptotics as taking both $N, p \rightarrow \infty$.
- ▶ Donohu (2006) shows that for $p > N$ case, when the true model is sparse, LASSO identified correct predictors with high probability (with certain assumptions on $\mathbf{X}$) as we slowly relax the penalty.
- ▶ Condition looks like (“good” variables are not too correlated with “bad” variables).

$$\max_{j \in S^c} \|x'_j X_S (X'_S X_S)^{-1}\|_1 \leq (1 - \epsilon)$$

- ▶ Where X_S are columns corresponding to nonzero coefficients, and S^c are set of columns with zero coefficients (at true value).

- ▶ *Grouped LASSO* for penalizing groups of coefficients at once (like fixed effects)
- ▶ *Relaxed LASSO* run LASSO to select coefficients and then run a non-penalized subset regression or LASSO with a less stringent penalty on the subset. (Here CV tends to pick a less strong penalty term λ leading to less shrinkage and bias).
- ▶ *SCAD: Smoothly Clipped Absolute Deviation*: do less shrinkage on big coefficients but preserve sparsity

$$\frac{dJ_a(|\beta|, \lambda)}{d\beta} = \lambda \cdot \text{sgn}(\beta) \left[I(|\beta| \leq \lambda) + \frac{(a\lambda - |\beta|)_+}{(a-1)\lambda} I(|\beta| > \lambda) \right]$$

- ▶ *Adaptive LASSO* uses a weighted penalty of the form $\sum_{j=1}^p w_j |\beta_j|$ where $W_j = 1/|\hat{\beta}_j|^\nu$ using the OLS estimates as weights. This yields consistent estimates of parameters while retaining the convexity property.

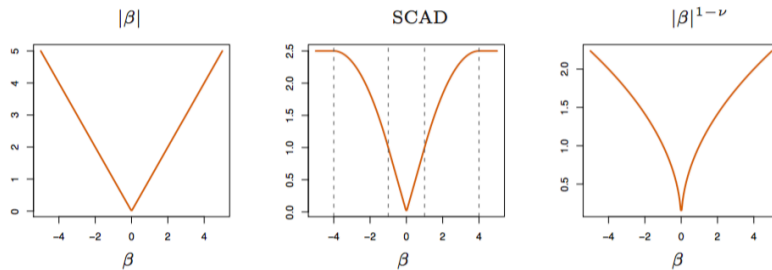


FIGURE 3.20. The lasso and two alternative non-convex penalties designed to penalize large coefficients less. For SCAD we use $\lambda = 1$ and $a = 4$, and $\nu = \frac{1}{2}$ in the last panel.

- ▶ Routines are highly specialized: there are lots of tricks
- ▶ No hope of coding this up on your own!
- ▶ In R you can use *glmnet* or *lars*.
- ▶ In Python you can use *scikit-learn*
- ▶ In most recent Matlab in Stats toolbox you have *lasso* and *ridge*.
- ▶ In STATA you can download *.ado* files from Christian Hansen's website (Chicago).